

Task 3

Task 1: Display the first 5 lines of /etc/group -> /file2.txt

```
head -n 5 /etc/group > /file2.txt
```

Task 2: Show first 10 lines of /root/anaconda-ks.cfg -> /anaconda.txt

```
head -n 10 /root/anaconda-ks.cfg > /anaconda.txt
```

Task 3: Display first 32 lines of /etc/passwd -> /file3.txt

```
head -n 32 /etc/passwd > /file3.txt
```

Task 4: Save last 10 lines of /etc/shadow -> /file4.txt

```
tail -n 10 /etc/shadow > /file4.txt
```

Task 5: Capture first 15 lines of /etc/passwd -> /file5.txt

```
head -n 15 /etc/passwd > /file5.txt
```

Task 6: Extract first 100 characters from /root/anaconda-ks.cfg -> /file6.txt

```
head -c 100 /root/anaconda-ks.cfg > /file6.txt
```

Task 7: Search lines ending with bash in /etc/passwd and save count -> /bash.txt

```
grep -c 'bash$' /etc/passwd > /bash.txt
```

Task 8: Lines containing nologin in /etc/passwd, count them -> /word.txt

```
grep -c 'nologin' /etc/passwd > /word.txt
```

Task 9: Find 'systemd' in both /etc/passwd and /etc/group -> /passgroup.txt

```
grep 'systemd' /etc/passwd /etc/group > /passgroup.txt
```

Task 10: Search words root, rhel, and systemd in /etc/passwd -> /search.txt

```
grep -E 'root|rhel|systemd' /etc/passwd > /search.txt
```

Task 11: Extract 5th column from /etc/passwd -> /file1.txt

```
cut -d: -f5 /etc/passwd > /file1.txt
```

```
# Task 12: Print columns 1 to 4 from /etc/passwd -> /file2.txt
```

```
cut -d: -f1-4 /etc/passwd > /file2.txt
```

```
# Task 13: Print columns 1 and 7 from /etc/passwd -> /file3.txt
```

```
cut -d: -f1,7 /etc/passwd > /file3.txt
```

```
# Task 14: Print 1st column from /etc/passwd, sort alphabetically -> /file4.txt
```

```
cut -d: -f1 /etc/passwd | sort > /file4.txt
```

```
# Task 15: Extract 3rd column from /etc/passwd -> /userid.txt
```

```
cut -d: -f3 /etc/passwd > /userid.txt
```

```
# Task 16: Awk print first and last columns from /etc/passwd -> /awkfile.txt
```

```
awk -F: '{print $1, $NF}' /etc/passwd > /awkfile.txt
```

```
# Task 17: Awk print cols 1, 2, 3 with delimiter '-' -> /awkfile2.txt
```

```
awk -F: -v OFS='-' '{print $1, $2, $3}' /etc/passwd > /awkfile2.txt
```

```
# Task 18: Search nologin in /etc/passwd, print first field -> /awkfile3.txt
```

```
awk -F: '/nologin/ {print $1}' /etc/passwd > /awkfile3.txt
```

```
root@shilpak:~  
[ec2-user@shilpak ~]$ sudo -i  
[root@shilpak ~]# head -n 5 /etc/group > /file2.txt  
[root@shilpak ~]# ls  
[root@shilpak ~]# exit  
logout  
[ec2-user@shilpak ~]$ ls  
Ramayan  auto.sh  auto.sh.save  course.txt  hardink-file  hello  linuxex  
[ec2-user@shilpak ~]$ sudo -i  
[root@shilpak ~]# cat /file2.txt  
root:x:0:  
bin:x:1:  
daemon:x:2:  
sys:x:3:  
adm:x:4:ec2-user  
[root@shilpak ~]#
```

```
[root@shilpak ~]# head -n 10 /root/anaconda-ks.cfg > /anaconda.txt  
[root@shilpak ~]# cat /anaconda.txt  
# Generated by Anaconda 34.25.0.29  
#version=RHEL9  
text  
repo --name="AppStream" --baseurl=http://mirror.centos.org  
%packages  
@^minimal-environment  
kexec-tools  
%end  
keyboard --xlayouts='us'  
lang en_US.UTF-8  
[root@shilpak ~]#
```

lang en_US.UTF-8

```
[root@shilpak ~]# head -n 32 /etc/pa
pam.d/ passwd passwd-
[root@shilpak ~]# head -n 32 /etc/passwd > /file3.txt
[root@shilpak ~]# cat /file3.txt
root:x:0:0:root:/root:/bin/bash
bin:x:1:1:bin:/bin:/sbin/nologin
daemon:x:2:2:daemon:/sbin:/sbin/nologin
adm:x:3:4:adm:/var/adm:/sbin/nologin
lp:x:4:7:lp:/var/spool/lpd:/sbin/nologin
sync:x:5:0:sync:/sbin:/bin/sync
shutdown:x:6:0:shutdown:/sbin:/sbin/shutdown
halt:x:7:0:halt:/sbin:/sbin/halt
mail:x:8:12:mail:/var/spool/mail:/sbin/nologin
operator:x:11:0:operator:/root:/sbin/nologin
games:x:12:100:games:/usr/games:/sbin/nologin
ftp:x:14:50:FTP User:/var/ftp:/sbin/nologin
nobody:x:65534:65534:Kernel Overflow User:/:/sbin/nologin
dbus:x:81:81:System message bus:/:/sbin/nologin
systemd-network:x:192:192:systemd Network Management:/:usr/sbin/nologin
systemd-oom:x:998:998:systemd Userspace OOM Killer:/:usr/sbin/nologin
systemd-resolve:x:193:193:systemd Resolver:/:usr/sbin/nologin
rpc:x:32:32:Rpcbind Daemon:/var/lib/rpcbind:/sbin/nologin
libstoragemgmt:x:997:997:daemon account for libstoragemgmt:/:usr/sbin/nologin
systemd-coredump:x:996:996:systemd Core Dumper:/:usr/sbin/nologin
systemd-timesync:x:995:995:systemd Time Synchronization:/:usr/sbin/nologin
sshd:x:74:74:Privilege-separated SSH:/usr/share/empty.sshd:/sbin/nologin
chrony:x:994:994:chrony system user:/var/lib/chrony:/sbin/nologin
ec2-instance-connect:x:993:993:/:home/ec2-instance-connect:/sbin/nologin
stapunpriv:x:159:159:systemtap unprivileged user:/var/lib/stapunpriv:/sbin/nologin
rpcuser:x:29:29:RPC Service User:/var/lib/nfs:/sbin/nologin
tcpdump:x:72:72:/:/sbin/nologin
ec2-user:x:1000:1000:EC2 Default User:/home/ec2-user:/bin/bash
jethalal:x:1001:1001:/:home/jethalal:/bin/bash
nginx:x:992:992:Nginx web server:/var/lib/nginx:/sbin/nologin
apache:x:48:48:Apache:/usr/share/httpd:/sbin/nologin
mysql:x:27:27:MySQL Server:/var/lib/mysql:/sbin/nologin
[root@shilpak ~]#
```

```
root@shilpak~  
[root@shilpak ~]#  
[root@shilpak ~]# tail -n 10 /etc/sh  
shadow shadow- shells  
[root@shilpak ~]# tail -n 10 /etc/shadow > /file4.txt  
[root@shilpak ~]# cat /file4.txt  
rpcuser:!!:20658:!!!!:  
tcpdump:!!:20658:!!!!:  
ec2-user:!!:20664:0:99999:7:::  
jethalal:!!:20669:0:99999:7:::  
nginx:!!:20683:!!!!:  
apache:!!:20684:!!!!:  
mysql:!!:20684:!!!!:  
student01:!!:20690:0:99999:7:::  
student02:$6$x9R.9l6pSTFTwapW$PaQLeUBK4oMgYNQ46t1Hsb0/2LamLpCwR.M8in4jbaIpswy1ARL7AkXR/sJSMfjESoecNuSyEJjvgbyK0YE2M.:20690:0:99999:7:::  
student03:!!:20690:0:99999:7:::  
[root@shilpak ~]#
```

```
[root@shilpak ~]# head -n 15 /etc/passwd > /file5.txt  
[root@shilpak ~]# cat /file5.txt  
root:x:0:0:root:/root:/bin/bash  
bin:x:1:1:bin:/bin:/sbin/nologin  
daemon:x:2:2:daemon:/sbin:/sbin/nologin  
adm:x:3:4:adm:/var/adm:/sbin/nologin  
lp:x:4:7:lp:/var/spool/lpd:/sbin/nologin  
sync:x:5:0:sync:/sbin:/bin/sync  
shutdown:x:6:0:shutdown:/sbin:/sbin/shutdown  
halt:x:7:0:halt:/sbin:/sbin/halt  
mail:x:8:12:mail:/var/spool/mail:/sbin/nologin  
operator:x:11:0:operator:/root:/sbin/nologin  
games:x:12:100:games:/usr/games:/sbin/nologin  
ftp:x:14:50:FTP User:/var/ftp:/sbin/nologin  
nobody:x:65534:65534:Kernel Overflow User:/:/sbin/nologin  
dbus:x:81:81:System message bus:/:/sbin/nologin  
systemd-network:x:192:192:systemd Network Management:/:/usr/sbin/nologin  
[root@shilpak ~]#
```

```

systemd-networkd:x:192:192:systemd Network Management:/:usr/sbin/nologin
[root@shilpak ~]# head -c 100 /root/anaconda-ks.cfg > /file6.txt
[root@shilpak ~]# cat /file6.txt
# Generated by Anaconda 34.25.0.29
#version=RHEL9
text
repo --name="AppStream" --baseurl=http://mirr[root@shilpak ~]# █

```

```

repo --name="AppStream" --baseurl=http://mirr[root@shilpak ~]#
[root@shilpak ~]# grep -c 'bash$' /etc/passwd > /bash.txt
[root@shilpak ~]# cat /bash.txt
6
[root@shilpak ~]# grep -c 'nologin' /etc/passwd > /word.txt
[root@shilpak ~]# cat /word.txt
26
[root@shilpak ~]# grep 'systemd' /etc/passwd /etc/group > /passgroup.txt
[root@shilpak ~]# cat /passgroup.txt
/etc/passwd:systemd-network:x:192:192:systemd Network Management:/:usr/sbin/nologin
/etc/passwd:systemd-oom:x:998:998:systemd Userspace OOM Killer:/:usr/sbin/nologin
/etc/passwd:systemd-resolve:x:193:193:systemd Resolver:/:usr/sbin/nologin
/etc/passwd:systemd-coredump:x:996:996:systemd Core Dumper:/:usr/sbin/nologin
/etc/passwd:systemd-timesync:x:995:995:systemd Time Synchronization:/:usr/sbin/nologin
/etc/group:systemd-journal:x:190:ec2-user
/etc/group:systemd-network:x:192:
/etc/group:systemd-oom:x:998:
/etc/group:systemd-resolve:x:193:
/etc/group:systemd-coredump:x:996:
/etc/group:systemd-timesync:x:995:
[root@shilpak ~]# grep -E 'root|rhel|systemd' /etc/passwd > /search.txt
[root@shilpak ~]# cat /search.txt
root:x:0:0:root:/root:/bin/bash
operator:x:11:0:operator:/root:/sbin/nologin
systemd-network:x:192:192:systemd Network Management:/:usr/sbin/nologin
systemd-oom:x:998:998:systemd Userspace OOM Killer:/:usr/sbin/nologin
systemd-resolve:x:193:193:systemd Resolver:/:usr/sbin/nologin
systemd-coredump:x:996:996:systemd Core Dumper:/:usr/sbin/nologin
systemd-timesync:x:995:995:systemd Time Synchronization:/:usr/sbin/nologin
[root@shilpak ~]# █

```

```
[root@shilpak ~]# cut -d: -f5 /etc/passwd > /file1.txt
[root@shilpak ~]# cat /file1.txt
root
bin
daemon
adm
lp
sync
shutdown
halt
mail
operator
games
FTP User
Kernel Overflow User
System message bus
systemd Network Management
systemd Userspace OOM Killer
systemd Resolver
Rpcbind Daemon
daemon account for libstoragemgmt
systemd Core Dumper
systemd Time Synchronization
Privilege-separated SSH
chrony system user

systemtap unprivileged user
RPC Service User

EC2 Default User

Nginx web server
Apache
MySQL Server

[root@shilpak ~]# █
```

```
root@shilpak ~]# cut -d: -f1-4 /etc/passwd > /file2.txt
[root@shilpak ~]# cat /file2.txt
root:x:0:0
bin:x:1:1
daemon:x:2:2
adm:x:3:4
lp:x:4:7
sync:x:5:0
shutdown:x:6:0
halt:x:7:0
mail:x:8:12
operator:x:11:0
games:x:12:100
ftp:x:14:50
nobody:x:65534:65534
dbus:x:81:81
systemd-network:x:192:192
systemd-oom:x:998:998
systemd-resolve:x:193:193
rpc:x:32:32
libstoragemgmt:x:997:997
systemd-coredump:x:996:996
systemd-timesync:x:995:995
sshd:x:74:74
chrony:x:994:994
ec2-instance-connect:x:993:993
stapunpriv:x:159:159
rpcuser:x:29:29
tcpdump:x:72:72
ec2-user:x:1000:1000
jethalal:x:1001:1001
nginx:x:992:992
apache:x:48:48
mysql:x:27:27
student01:x:1002:1004
student02:x:1003:1005
student03:x:1004:1007
[root@shilpak ~]#
```

root@shilpak~

```
[root@shilpak ~]# cut -d: -f1,7 /etc/passwd > /file3.txt
```

```
[root@shilpak ~]# cat /file3.txt
```

root:/bin/bash

bin:/sbin/nologin

daemon:/sbin/nologin

adm:/sbin/nologin

lp:/sbin/nologin

sync:/bin/sync

shutdown:/sbin/shutdown

halt:/sbin/halt

mail:/sbin/nologin

operator:/sbin/nologin

games:/sbin/nologin

ftp:/sbin/nologin

nobody:/sbin/nologin

dbus:/sbin/nologin

systemd-network:/usr/sbin/nologin

systemd-oom:/usr/sbin/nologin

systemd-resolve:/usr/sbin/nologin

rpc:/sbin/nologin

libstoragemgmt:/usr/sbin/nologin

systemd-coredump:/usr/sbin/nologin

systemd-timesync:/usr/sbin/nologin

sshd:/sbin/nologin

chrony:/sbin/nologin

ec2-instance-connect:/sbin/nologin

stapunpriv:/sbin/nologin

rpcuser:/sbin/nologin

tcpdump:/sbin/nologin

ec2-user:/bin/bash

jethalal:/bin/bash

nginx:/sbin/nologin

apache:/sbin/nologin

mysql:/sbin/nologin

student01:/bin/bash

student02:/bin/bash

student03:/bin/bash

```
[root@shilpak ~]#
```

root@shilpak:~

```
[root@shilpak ~]# cut -d: -f1 /etc/passwd | sort > /file4.txt
```

```
[root@shilpak ~]# cat /file4.txt
```

adm

apache

bin

chrony

daemon

dbus

ec2-instance-connect

ec2-user

ftp

games

halt

jethalal

libstoragemgmt

lp

mail

mysql

nginx

nobody

operator

root

rpc

rpcuser

shutdown

sshd

stapunpriv

student01

student02

student03

sync

systemd-coredump

systemd-network

systemd-oom

systemd-resolve

systemd-timesync

tcpdump

```
[root@shilpak ~]#
```

root@shilpak:~

```
[root@shilpak ~]# cut -d: -f3 /etc/passwd > /userid.txt
```

```
[root@shilpak ~]# cat /userid.txt
```

0

1

2

3

4

5

6

7

8

11

12

14

65534

81

192

998

193

32

997

996

995

74

994

993

159

29

72

1000

1001

992

48

27

1002

1003

1004

```
[root@shilpak ~]#
```

root@shilpak:~

```
[root@shilpak ~]# awk -F: '{print $1, $NF}' /etc/passwd > /awkfile.txt
```

```
[root@shilpak ~]# cat /awkfile.txt
```

```
root /bin/bash
bin /sbin/nologin
daemon /sbin/nologin
adm /sbin/nologin
lp /sbin/nologin
sync /bin/sync
shutdown /sbin/shutdown
halt /sbin/halt
mail /sbin/nologin
operator /sbin/nologin
games /sbin/nologin
ftp /sbin/nologin
nobody /sbin/nologin
dbus /sbin/nologin
systemd-network /usr/sbin/nologin
systemd-oom /usr/sbin/nologin
systemd-resolve /usr/sbin/nologin
rpc /sbin/nologin
libstoragemgmt /usr/sbin/nologin
systemd-coredump /usr/sbin/nologin
systemd-timesync /usr/sbin/nologin
sshd /sbin/nologin
chrony /sbin/nologin
ec2-instance-connect /sbin/nologin
staprunpriv /sbin/nologin
rpcuser /sbin/nologin
tcpdump /sbin/nologin
ec2-user /bin/bash
jethalal /bin/bash
nginx /sbin/nologin
apache /sbin/nologin
mysql /sbin/nologin
student01 /bin/bash
student02 /bin/bash
student03 /bin/bash
[root@shilpak ~]#
```

```
root@shilpak:~  
[root@shilpak ~]# awk -F: -v OFS='-' '{print $1, $2, $3}' /etc/passwd > /awkfile2.txt  
[root@shilpak ~]# cat /awkfile2.txt  
root-x-0  
bin-x-1  
daemon-x-2  
adm-x-3  
lp-x-4  
sync-x-5  
shutdown-x-6  
halt-x-7  
mail-x-8  
operator-x-11  
games-x-12  
ftp-x-14  
nobody-x-65534  
dbus-x-81  
systemd-network-x-192  
systemd-oom-x-998  
systemd-resolve-x-193  
rpc-x-32  
libstoragemgmt-x-997  
systemd-coredump-x-996  
systemd-timesync-x-995  
sshd-x-74  
chrony-x-994  
ec2-instance-connect-x-993  
staprunpriv-x-159  
rpcuser-x-29  
tcpdump-x-72  
ec2-user-x-1000  
jethalal-x-1001  
nginx-x-992  
apache-x-48  
mysql-x-27  
student01-x-1002  
student02-x-1003  
student03-x-1004  
[root@shilpak ~]#
```

```
root@shilpak:~  
[root@shilpak ~]# awk -F: '/nologin/ {print $1}' /etc/passwd > /awkfile3.txt  
[root@shilpak ~]# cat /awkfile3.txt  
bin  
daemon  
adm  
lp  
mail  
operator  
games  
ftp  
nobody  
dbus  
systemd-network  
systemd-oom  
systemd-resolve  
rpc  
libstoragemgmt  
systemd-coredump  
systemd-timesync  
sshd  
chrony  
ec2-instance-connect  
staprunpriv  
rpcuser  
tcpdump  
nginx  
apache  
mysql  
[root@shilpak ~]#
```